CHAPTER 16

The Social Implications of Advanced Health Care Technology

Learning Objectives

- Identify five key recent advancements in medical technology.
- Identify and discuss five important social implications of rapidly developing medical technology.
- Explain the key arguments in support of and opposed to patients being able to demand and to refuse medical treatment. Identify and explain key arguments supporting and opposing the legality of physician-assisted suicide.
- Discuss important issues related to organ donation. Discuss important issues related to organ donation policy.
- Identify and describe four modern assisted procreative techniques. Identify and explain key arguments supporting and opposing the legality of surrogate motherhood.

The development of **technology**—the practical application of scientific or other forms of knowledge—is a major stimulus of social change in most modern societies. Western cultures subscribe to a belief system that prioritizes "technical rationality"—a mind-set that "essentially all problems are seen as manageable with technical solutions, and rationality (reasonableness, plausibility, proof) can be established only through scientific means using scientific criteria" (Barger-Lux and Heaney, 1986:1314). However, many social scientists believe that technology not only is influenced by cultural values but also in return has a powerful and deterministic effect on culture and social structure—a theory known as **technological determinism.**

Today's health care system reflects the rapid rate of technological innovation in the last few decades. Hospitals and medical clinics contain sophisticated pieces of equipment and specially trained personnel to operate them. The benefits of advanced health care technologies are apparent: more accurate and quicker diagnoses, effective treatment modalities, and increased life expectancy. However, there are also negative consequences of technological innovations, including increased costs, inequities of access, technological "advancements" that fail (e.g., the artificial heart and thalidomide), and troubling ethical issues (Chang and Lauderdale, 2009).

SOCIETAL CONTROL OF TECHNOLOGY

Advocates view technological development as a means for society to fulfill its needs and to create a better life for its citizens. The need for more powerful means of information storage and processing produced the computer revolution. The need for faster food preparation techniques for on-the-go families led to the microwave oven. Automobile air bags are a safety innovation in a society where thousands lose their lives each year in traffic accidents. According to this view (sometimes referred to as a *utopian* view), society controls the introduction of new technologies; technological advancements continue because they are beneficial to society.

Others, however, are concerned that technologies also create problems (a *dystopian* view). They critique modern societies (especially the United States) for a failure to systematically assess potential technologies in order to determine whether or not they should be pursued. Instead, American society is said to be controlled by a **technological imperative**—the idea that "if we have the technological capability to do something, then we should do it. . . . [it] implies that action in the form of the use of an available technology is always preferable to inaction" (Freund and McGuire, 1999:243).

Critics charge that this technological imperative is clearly demonstrated in medicine: in the desire of individual physicians to do the newest and most sophisticated procedures, even if more conservative treatment would be as appropriate; in health insurance companies' greater willingness to pay for high-tech medicine rather than low-tech or nontech care; and in the march of hospitals to create (and thus be forced to use) high-tech wards (such as coronary intensive care units), even when they are shown not to offer any consistent advantage over more conservative, low-tech, less expensive forms of treatment (Barger-Lux and Heaney, 1986; Freund and McGuire, 1999). Historian David Rothman argues that the insistence of the middle class to unfettered access to medical technologies has been the most important influence on America's health policy for at least the last 60 years (Rothman, 1997).

HEALTH CARE TECHNOLOGY

Advancements in health care technology occurred throughout the last century and the first part of this one, but the pace of development in the last few decades has been phenomenal. Bronzino, Smith, and Wade (1990) identify the following key advancements during these years:

1. ***Cardiac technology.*** Important innovations include the cardiac pacemaker, which senses the heart's own electrical activity and paces it appropriately; the defibrillator, which maintains the rhythmic contractions of the heart to avoid a "heart attack"; and heart transplants.
2. ***Critical care medicine.*** Significant advances have been made in handling intensive care

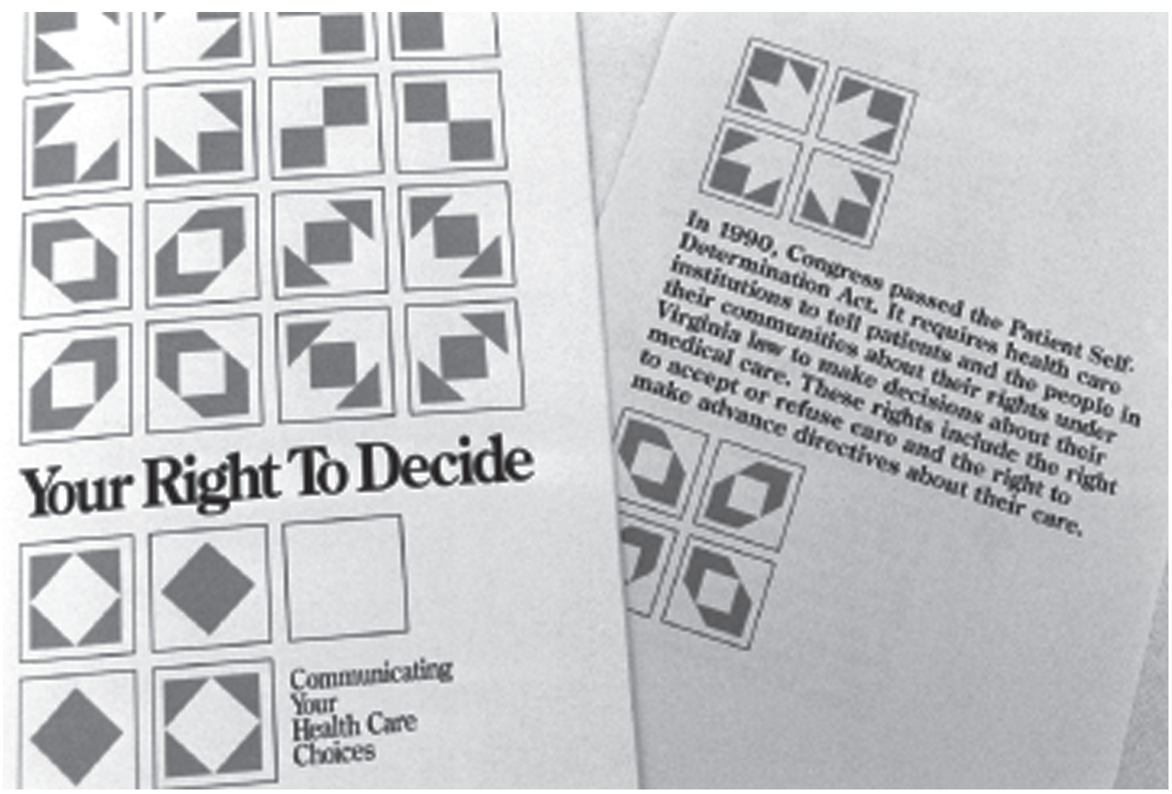
Magnetic resonance imaging, which uses magnetic fields, radio wave energy, and sophisticated computer software, has enabled clearer and more detailed pictures of internal organs. MRI machines can cost up to $3 million, and the average price per MRI exceeds $3,500.

unit (ICU) cardiopulmonary patients (those with insufficient heart and lung capacity). An estimated 20 percent of all hospital patients require some form of respiratory therapy or support, including administration of oxygen to patients who cannot maintain adequate oxygen levels in their blood with their own breathing; performance of physical therapy to break up secretions and mucus in the lungs; and mechanical ventilation for patients unable to breathe on their own.

3. ***Medical imaging.*** Noninvasive techniques such as nuclear medicine, ultrasound, computer tomography (CT; also called computerized axial tomography—CAT), and magnetic resonance imaging (MRI) allow pictures to be taken of internal bodily organs. Recent advances provide even more information about bodily tissues.
4. ***Health care computers (information technology).*** These are used throughout the modern health care facility—in the clinical laboratory, in instrumentation, in building patient databases, and in diagnostic support systems. Some analysts believe that in the near future tens of millions of Americans will wear wireless monitoring devices that automatically send vital signs to medical professionals—an extension of devices that now provide automatic fall detection. See the accompanying box, "Telemedicine."

In addition, we are in the early stages of at least a fifth advancement:

5. ***Genomic medicine.*** While genetics examines single genes and their function, **genomics** examines the interaction of multiple genes together and in interaction with the environment. Many diseases including breast and colorectal cancers, HIV/AIDS, Parkinson's, and Alzheimer's can best be understood and addressed with this multifactorial approach. This knowledge is also leading to personalized medicine in which genetic tests determine the specific treatment that will be used for individual patients (Liotta and Petricoin, 2008).



Many hospitals now provide patients with a pamphlet that describes their right to compose a living will and/or to authorize a health care power of attorney. The Patient Self-Determination Act requires that hospitals make these options known to patients.

The Social Implications of Advanced Health Care Technology

Sociologists and other social scientists have identified at least five specific social implications of advanced health care technologies.

First, advanced health care technologies create options for people and society. These include using today's sophisticated emergency personnel and equipment to sustain a life that once would have expired, cardiac bypass surgery (where blocked cardiac arteries can be replaced by veins taken from the leg), and assisted procreation.

Second, advanced health care technologies alter human relationships. The existence of technological apparatuses that are able to sustain life after consciousness has been permanently lost has caused families throughout the country to discuss their personal wishes and have created difficult decisions for family members of individuals whose wishes are unknown. Physicians and other health care professionals consider the option of "DNR" ("do not resuscitate") or "no code"—and engage family members in discussions about it.

IN THE FIELD

TELEMEDICINE

The information superhighway has created many new opportunities for sharing, obtaining, and discussing information. While development and use of this technology in medicine (as in other fields) is still in its infancy, already some of the potential is evident. Research has found that more than 85 percent of physicians and 66 percent of adults in the United States use the Internet for health-related information retrieval (Parekh, Mayer, and Rjowsky, 2009). In one survey, about half of female respondents and almost 40 percent of male respondents said they check online for medical information before they contact a physician (Harris Interactive Poll, 2008). A dozen states have passed legislation that requires private insurers to cover services that are provided through telemedicine. Medicaid already compensates for telemedicine services in most states and will soon do so in all 50 states. More than one-third of physicians now e-mail patients, although 90 percent of patients would like them to do so. Following are some ways that the Internet is affecting health care:

- Some dot-com physicians ("cyberdocs") have established Web sites that patients (sometimes referred to as "guests") can contact and chat one-on-one with available physicians; these sessions are viewed as an alternative to an office visit; many physicians remain skeptical.
- Home health care nurses can "virtually" visit patients through monitors that enable the two to see each other while talking on the telephone and even to check the patient's heart rate (the patient uses a stethoscope; the nurse uses a headset attached to the computer). This is being referred to as "telehealth." More than a dozen states now allow doctors to treat patients and to prescribe drugs online without any personal contact.
- Anyone with Internet access can retrieve information about any health care condition and treatment from a variety of Web sources. This reduces the traditional dominance of the physician as gatekeeper to medical knowledge and enables individuals to directly access information. However, because well-educated middle- and upper-income individuals have the most Internet access, they will continue to obtain the most information.
- An estimated 40,000 to 50,000 smartphone health care applications are now available and they were downloaded an estimated 247 million times in 2012—a year in which the number of health app users doubled. The Food and Drug Administration predicts 500 million worldwide users in 2015. Especially popular is Epocrates, a drug reference tool with more than 330,000 active physician users, followed by UpToDate and

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Medscape—both clinical decision support reference tools.

- The reading of diagnostic tests is now sometimes "outsourced"—that is, it is read and interpreted by a physician in another country. For example, if a patient in the United States undergoes an emergency brain scan in the middle of the night, it will be electronically sent to a physician perhaps in India or Australia for interpretation rather than calling a radiologist into the hospital—this is referred to as "teleradiology."
- Electronic files of patient data can be created, stored, and easily communicated to other involved health care professionals. Many analysts believe there is significant cost-saving potential through these electronic health records (EHRs) (Shea and Hripcsak, 2010). In 2013 the federal government announced that more than 50 percent of all physicians were using electronic health records.
- Medical journals can place abstracts or full text of articles on the Internet.
- Health education online games are being created for children and adolescents to teach about such things as healthy lifestyles and the importance of taking prescribed medications. Public health departments are using tools such as Facebook, texting, and twittering to provide easily available information to young people.
- Electronic support groups ("e-health") are now available for thousands of diseases and conditions and millions of Americans. Research has identified benefits for users and especially for those who create health information through blogging and contributing to social networking on health topics (Ziebland and Wyke, 2012).
- Electronic mail can be used to enhance communication—especially among those located in rural areas without a wide support network.
- Online physician "report cards" are now available enabling patient evaluations of medical care providers.
- Educational courses can be offered on the Internet—as is already occurring in efforts to bring more public health information to health professionals and patients in Third World countries.

Assessment of these interventions is still in the early stages, but promising results have been found. However, it is clear that there is a wide range in the accuracy and quality of online health information, so consumers must remain very careful in selecting reputable sites (Goldsmith, 2000; Wachter, 2006).

Many people are concerned that the continuing introduction of advanced health care technologies has led to a dehumanization of patient care. Patients complain that physicians concentrate so much on the disease (in anticipation of selecting and using the appropriate technology) that they lose sight of the patient as a person. Feeling that they are being treated more like a thing than a person, patients despair that the warmth and empathy demonstrated by many physicians in the pre-high-tech era are being lost (Barger-Lux and Heaney, 1986). Discussion today of "cyberdocs," "robodocs," and "virtual doctors" indicates the increasing use of modern computer wizardry in medicine but suggests to many an increasingly distant physician–patient relationship.

Renee Anspach's ethnographic work *Deciding Who Lives: Fateful Choices in the Intensive-Care Nursery* (1993) illustrates this concern. Anspach discovered that the different ways in which physicians and nurses relate to infants in an intensive care (IC) nursery affect their medical interpretations. Physicians, who have more limited contact with the infants and whose interaction is primarily technologically focused, rely on diagnostic technology to develop prognoses. Nurses, whose contact with the infants is more continuous, long-term, and emotional, develop prognoses more on the basis of their interaction and observations. While Anspach does not suggest that either prognostic technique is superior, her analysis demonstrates that one's position in the social structure of the nursery influences perceptions and that, to a certain extent, technology may "distance" the physician from the patient.

Third, advanced health care technologies affect the entire health care system. For example, technology has been the most important stimulus for the rapid increase in health care costs. An estimated half of recent health care inflation is due to new technologies. Much of the problem relates to increasingly expensive equipment like CT scanners and nuclear medicine cameras, which cost upward of $500,000. The overall price tag for medical technologies in the world's developed countries in 2010 approached $1 trillion.

The United States is now confronting the realization that it cannot afford every potentially helpful medical procedure for every patient. Increasing the amount of funds spent on health care (already considered by many to be at an unacceptably high level) would mean reducing the amount of money spent on education, the environment, and/or other areas of government funding. When the country (i.e., the government or health insurance companies) chooses to subsidize new technologies, it is explicitly or implicitly choosing not to subsidize other health care programs.

These macro-allocation decisions have led to increasingly sophisticated means of technology assessment and cost-benefit analysis. Efforts have been undertaken to quantify the outcomes of implementing specific technologies and compare these outcomes with those expected from other health care programs. For example, should the government fund 50 organ transplant procedures or offer prenatal care to 5,000 low-income women? The accompanying box, "Technology Assessment in Medicine," discusses this process.

These decisions also include value questions related to such issues as the amount of money spent on preventive care versus curative measures, the amount of money spent on people near the end of their lives, the amount of money spent on newborns who will require extensive lifetime care, the amount of money spent on diseases related to "voluntary lifestyles," and mechanisms to provide equal access to available programs.

Fourth, advanced health care technologies stimulate value clarification thinking. Medicine continues to raise issues that force individuals to confront provocative value questions about life and death.

As discussed in Chapter 4, the United States and other countries have recently mapped and sequenced the human genome (the complete set of human genes). The knowledge gained is dramatically increasing our understanding of human evolution and humans' genetic relatedness with other organisms, the connection between genes and human behavior, and the relationship between specific genes and particular diseases. It is conceivable that this knowledge eventually will enable the elimination or control of all genetic diseases.

However, this process also raises value questions. For example, if all genetic diseases can someday be diagnosed and eliminated during fetal development, will such action be required? Would broad genetic screening programs with mandatory participation for all prospective parents or pregnant women be established? Would it be illegal or immoral to produce a child with an unnecessary genetic disease? Could such a child file a "wrongful life" suit against her or his parents?

Who should (or likely would) have access to genetic information about individuals? Would an employer have a right to a genetics background check of prospective employees? Would health insurance companies be able to require a genetics check-up when deciding whether or not to offer insurance to someone or in calculating the cost of the insurance policy? How would discrimination in hiring or insuring be proven and be handled?

In addition to doing prenatal therapy to correct defective genes, will it be permissible to do therapy to provide for genetic enhancements? For example, would it be legal to attempt to "boost" the intelligence gene(s), or would it be ethical to abort any (or even all?) fetuses with low intellectual potential? Should the government subsidize prospective parents who

IN THE FIELD

TECHNOLOGY ASSESSMENT IN MEDICINE

What is health technology assessment (HTA)? According to Lehoux and Blume (2000:1063), it is "a field of applied research that seeks to gather and synthesize the *best available evidence* on the costs, efficacy, and safety of health technology." Littenberg (1992:425–427) recommends that any proposed medical technology be assessed on these five levels:

1. *Biologic plausibility* assesses whether "the current understanding of the biology and pathology of the disease in question can support the technology."
2. *Technical feasibility* assesses whether "we can safely and reliably deliver the technology to the target patients."
3. *Intermediate outcomes* assess the immediate and specific "biological, physiologic, or clinical effects of the technology."
4. *Patient outcomes* assess "overall and ultimate outcomes for the health of the patient."
5. *Societal outcomes* assess the external effects of the technology on society, including the ethical and fiscal consequences.

A recent example demonstrating assessment of technology impact is use of surgical robots. They have received significant attention in the popular press and have become very popular in hospital operating rooms and among hospital marketing staffs. By 2012 more than 1,500 surgical robots were in use in the United States, and worldwide, more than 360,000 procedures were robot-assisted. They are, of course, very expensive. Typical cost is $2.3 million each with $135,000 additional annually required for service fees. Using surgical robots adds an estimated $3,000 to $6,000 to each surgery.

Assessment, however, has determined that there is no clear benefit of surgical robots over traditional, human hands-on surgery with regard to patient outcome (robotic surgery does require smaller incisions). Perhaps additional benefit will occur as the devices evolve, but thus far they seem to be mostly driven by the technological imperative.

want therapy to produce a taller offspring or a shorter one? What are the long-term implications for altering the gene pool? These questions exemplify the difficult value questions that are created by this new knowledge.

Finally, advanced health care technologies create social policy questions. Of course, issues that raise difficult value questions for individuals often raise complicated social policy questions for societies. Critics have charged that legislatures and courts have shaped policy regarding ethical issues in medicine prior to adequate public debate. This is beginning to change as these issues are now receiving greater public scrutiny.

An example of an advanced medical technology forcing social policy consideration occurs with **fetal tissue transplants.** The vast majority of abortions are performed in the first trimester by the suction curettage method in which the fetus is removed by suction through a vacuum cannula. While the fetus is fragmented in this procedure, cells within tissue fragments can be collected. In about 10 percent of abortions using suction curettage (about 90,000 per year), the fragment containing the fetal midbrain can be identified and retrieved.

Research has demonstrated that transplantation of an aborted fetus's midbrain can relieve or even cure certain diseases, including diabetes, Parkinson's disease, and Alzheimer's disease. A high rate of successful transfer is due partially to the fact that this very immature tissue is unlikely to be rejected by the recipient. In the early 2000s the issue became focused on the retrieval of stem cell—a universal cell residing in embryos and fetuses where it is called upon to construct hearts, lungs, brains, and other vital organs and tissues. When retrieved, stem cells can be transformed or molded into any type of organ or tissue. The created organ or tissue would potentially be a cure for a wide range of diseases.

However, fetal tissue and stem cell transplants are opposed by most of those who believe that personhood begins at the instant of conception. Because the fetus or embryo is destroyed in the process of obtaining the tissue or stem cells, it is believed that the procedure (even for beneficial result) is an act of complicity in the ending of a human life. Presidents Ronald Reagan and George H.W. Bush placed a moratorium on publicly sponsored research on fetal tissue transplants. President Clinton overturned this ban. President George W. Bush enacted a policy that allowed research on a relatively small number of stem cell lines already in existence but prohibited public support for research on additional lines, thus restricting and reducing research in this area. President Obama issued an executive order restoring publicly funded research.

Social Issues Raised by Advancing Health Care Technology

The remainder of this chapter examines three issues that raise questions about the rights of individuals and patients versus the force of technology. Each issue has led to serious discussion of the rights of individuals to access or refuse to access modern medical technology and of the responsibilities of society to control and regulate use of the technology. These questions involving individual choice versus public good and the means by which medical resources are allocated have become increasingly common as sophisticated high-technology care continues to be developed and made available.

Do patients have a legal right to refuse medical treatment—including artificial means of nourishment? Can patients demand a particular medical treatment even if physicians judge it to be futile? Can physicians participate in patient suicides? Should organ donation policies be revised to obtain more organs for transplantation? Should there be more investigation of the way that human relationships are altered by decisions to give and receive an organ? Should assisted procreative techniques be made available to anyone who wants them? Is it proper for society to regulate these techniques and/or even to prohibit them?

THE RIGHT TO REFUSE OR DEMAND ADVANCED HEALTH CARE TECHNOLOGY

Health care technology is at a stage of development in which it is often able to keep people alive but without being able to cure their disease, relieve their pain, or, at times, even restore consciousness. As this critical care technology (especially the artificial ventilator) has been developed and incorporated in hospitals, the customary practice has been to use it whenever possible.

Gradually, however, patients and their families have begun to challenge the unquestioned use of technology and have asked (in the words of the book, play, and film) "Whose life is it, anyway?" Patients and/or their proxies have become more assertive in requesting, and sometimes demanding, that the technology be withheld or withdrawn. In some circumstances, physicians and hospitals have complied, but, in other circumstances, requests have been refused. These situations have often ended in court hearings, and it has been the court system rather than legislators that has primarily dealt with the rights of patients/families versus the rights of hospitals/physicians to determine the use of, or refusal to use, advanced health care technologies.

Do Patients Have a Legal Right to Refuse Medical Treatment?

Do competent patients, and incompetent patients through their representatives, have a right to refuse medical treatment? Or, are physicians and hospitals required (or, at least, lawfully able) to use all forms of medical treatment, including high-tech medicine, whenever they deem that to be appropriate? While this issue has been discussed for many years, three landmark court cases have addressed this issue of "fundamental rights."

Karen Ann Quinlan. In April 1975, 21-year-old **Karen Ann Quinlan** was brought to a hospital emergency room. She had passed out at a party and temporarily stopped breathing (during which time part of her brain died from lack of oxygen). Blood and urine tests showed that she had had only a couple of drinks and a small amount of aspirin and Valium, but that significant brain damage had occurred. Karen was connected to an artificial ventilator to enable respiration (see the box, "Defining Death").

After four months, the Quinlans acknowledged that Karen was unlikely ever to regain consciousness, and that she would be severely brain damaged if she did. Their priest assured them that the Catholic Church did not require continuation of extraordinary measures to support a hopeless life. The family asked that the artificial ventilator be disconnected. The hospital refused, arguing that Karen was alive and that it was their moral and legal obligation to act to sustain her life. The Quinlans went to court asking to be designated as Karen's legal guardians in order that the ventilator could be disconnected. Part of the rationale offered by their attorney was that the Constitution contains an implicit right to privacy that guarantees that individuals (or people acting on their behalf) can terminate extraordinary medical measures even if death results.

The Karen Ann Quinlan case of 1970s raised extremely difficult questions for her family—shown here—and for society and the medical profession about the right to discontinue medical treatment.

The Superior Court ruled against the Quinlan family, but the case was appealed to the New Jersey Supreme Court, which overruled the prior decision and granted guardianship to Mr. Quinlan. The court ruled that patients have a constitutionally derived right to privacy that includes the right to refuse medical treatment, and this right extends to competent and incompetent persons.

After a protracted series of events, the respirator was disconnected. When this occurred, surprisingly, Karen began to breathe on her own. It was determined that she was in a **persistent vegetative state (PVS).** In PVS, the patient is not conscious, is irretrievably comatose, is nourished artificially, *but* is respirating on his or her own. This happens when the brain stem is functioning, but the cerebrum is not. The eyes are open at times; there are sleep–wake cycles; the pupils respond to light; and gag and cough reflexes are normal. However, the person is completely unconscious and totally unaware of surroundings and will remain in this state till death (which may not occur for many years). At any one time, about 10,000 people in the United States are in PVS. It is not the same as **brain death**—in which neither the cerebrum nor the brain stem is functioning. To be sustained, patients require artificial nutrition and hydration but no other forms of medical treatment. To be clear, the court ruled about medical treatment

IN THE FIELD

DEFINING DEATH

The brain consists of three divisions:

1. The *cerebrum* (with the outer shell called the cortex; also called the "higher brain") is the primary center of consciousness, thought, memory, and feeling; many people believe it is the key to what makes us human—that is, it establishes "personhood."
2. The *brain stem* (also called the "lower brain") is the center of respiration and controls spontaneous, vegetative functions such as swallowing, yawning, and sleep–wake cycles.
3. The *cerebellum* coordinates muscular movement.

Historically, death was defined as the total stoppage of respiration and pulsation. Any destruction of the brain stem would stop respiration, denying needed oxygen to the heart, which would stop pulsation; death would typically occur within 20 minutes.

This definition was rendered inappropriate by the artificial respirator, which, in essence, replaces the brain stem. It enables breathing and therefore heartbeat.

In 1968, the brain death definition of death was developed at Harvard Medical School. It defines death as a permanently nonfunctioning whole brain (cerebrum and brain stem) including no reflexes, no spontaneous breathing, no cerebral function, and no awareness of externally applied stimuli. Thus, if breathing persists, but only through means of an artificial respirator, the person is officially dead. Many persons today would prefer a higher brain-oriented definition of death like that suggested by Robert Veatch (1993:23), "an irreversible cessation of the capacity for consciousness." One effect of this definition would be that patients in a persistent (or "permanent") vegetative state would be declared to be dead.

in the Quinlan case—not about nutrition and hydration. Karen was moved to a chronic care institution, where she continued breathing for ten years before expiring.

Nancy Cruzan. Does this right to privacy extend to the refusal to accept artificial means of nourishment? (This is typically done through a nasogastric tube that delivers fluids through the nose and esophagus, or a gastrostomy in which fluids are delivered by tube through a surgical incision directly into the stomach, or intravenous feeding and hydration in which fluids are delivered through a needle directly into the bloodstream.) Or, is the provision of nutrition and hydration so basic that it is not considered to be "medical treatment"?

In a poignant and lengthy travail through the court system, the case of **Nancy Cruzan** provided a judicial answer to the question. In January 1983, Nancy was in an automobile accident and suffered irreversible brain damage. She entered a PVS. After four years, Nancy's parents asked the Missouri Rehabilitation Center to withdraw the feeding tube. The center refused; the Cruzans filed suit. They informed the court that Nancy had indicated that she would not want to be kept alive unless she could live "halfway normally." The Circuit Court ruled in favor of the Cruzans, but under appeal, the Missouri Supreme Court overruled—disallowing the feeding tube from being disconnected. The court concluded that there was not "clear and convincing evidence" that Nancy would not have wanted to be maintained as she was. Missouri law required such evidence for the withdrawal of artificial life-support systems.

The Cruzans appealed to the U.S. Supreme Court, and, for the first time, the Court agreed to hear a "right to die" case. On June 25, 1990, the Court handed down a 5–4 decision in favor of the state of Missouri. States were given latitude to require "clear and convincing evidence" that the individual would not wish to be sustained in

IN THE FIELD

ADVANCE DIRECTIVES

A **living will** is a document signed by a competent person that provides explicit instructions about desired end-of-life treatment if the person is unconscious or unable to express his or her wishes. Legal in all 50 states and the District of Columbia, the living will is commonly used to authorize the withholding or withdrawal of life-sustaining technology and provides immunity to health care professionals who comply with the stated wishes. Approximately 30 percent of adult Americans have completed a living will. Ironically, in most situations, physicians and hospitals ignore the living will if family members request medical treatment.

A *health care power of attorney* can be signed by a competent person to designate someone who will make all health care decisions should the person become legally incompetent. It requires the person to be available in the needed situation but allows the person to consider the particulars of the situation before making a decision. Most experts agree that both types of advance directives have some benefit and encourage people to do both.

The *Patient Self-Determination Act* (sometimes called the "medical Miranda warning") went into effect on December 1, 1991. It requires all health care providers who receive federal funding to inform incoming patients of their rights under state laws to refuse medical treatment and to prepare an advance directive. The law is intended to ensure that people are aware of their rights vis-à-vis life-sustaining medical technology.

PVS—meaning that families could not use their own judgment to decide to discontinue feeding.

However, the Court did acknowledge a constitutional basis for **advance directives** (see the accompanying box "Advance Directives") and the legitimacy of surrogate decision making. Its rationale cited the Fourteenth Amendment's "liberty interest" as enabling individuals to reject unwanted medical treatment. Because the decision did not distinguish between the provision of nutrition and other forms of medical treatment, the feeding tubes could have been removed had the qualifications of the Missouri law been met.

Ultimately, another hearing occurred in the Circuit Court; friends of Nancy came forward to offer additional evidence that Nancy would not have wanted to be maintained in a PVS; and the judge again ruled for the Cruzans. No appeal was filed, and the feeding tubes were removed on December 14, 1990. Nancy died 12 days later.

Terri Schiavo. In 2004 and 2005, the nation's attention was riveted on a Florida case that once again raised issues regarding persistent vegetative states, the role of proxy decision makers, and the power of the media and politicians to dramatize complex medical issues. On February 25, 1990, 26-year-old **Terri Schiavo** suffered a cardiac arrest that was, at least in part, brought on by an eating disorder. The disruption of oxygen caused permanent brain damage. Like Nancy Cruzan (and ultimately, Karen Ann Quinlan), Terri entered a PVS and was kept alive by artificial nutrition and hydration.

For several years, her case was not unlike that of thousands of other PVS patients as both her husband, Michael, and her birth family did all they could to provide support. In 1994, the relationship between Michael and her parents and siblings broke down over the manner in which a medical malpractice judgment would be spent. Four years later, in 1998, having become resigned to the fact that Terri's brain damage was permanent, Michael asked that the feeding tubes be withdrawn. Like Nancy Cruzan, Terri did not have any form of advance directive, and Michael was the legal guardian and proxy.

Her family opposed this decision and filed a series of court cases over the next several years to remove Michael as guardian. All found that Michael was acting within his legal rights. In the early 2000s, the media picked up on the case, and details of the situation were reported on a routine basis. Governor Jeb Bush, the Florida state legislature, and eventually, President Bush and the Republican leadership in Congress became involved in the case and sought all available means to overturn the several court rulings in the case and to prevent the withdrawal of the feeding tube. Twice, the feeding tube was withdrawn and then reinserted. Congressional Republicans passed a bill that pertained just to Terri, but federal judges immediately found it to be unconstitutional. Ultimately, every court involved in the case ruled in favor of Michael's right to decide. The feeding tube was once again removed, and Terri died (Hampson and Emanuel, 2005; Werth, 2006). Political involvement and partisan wrangling in the case was at such a high level, according to the results of one study conducted in the aftermath of *Schiavo*, that college students began seeing this issue more from a political than ethical viewpoint (Weiss and Lupkin, 2009–2010).

Can Patients Demand a Particular Medical Treatment?

The cases of Karen Ann Quinlan, Nancy Cruzan, and Terri Schiavo illustrate circumstances in which a patient or her or his proxy wishes to refuse medical treatment. In 1991, an unusual twist occurred in the Helga Wanglie case: The hospital wished to stop medical treatment, and the family demanded that it continue. This created the reverse of the usual question: Does a patient or her or his representatives have the right to demand medical treatment?

Helga Wanglie

In December 1989, **Helga Wanglie,** an active, well-educated 85-year-old woman, tripped over a rug and broke her hip. During the next several months, she experienced several cardiopulmonary arrests. In May 1990, she suffered severe anoxia and slipped into a PVS. Her breathing was reinforced by a respirator, and she was fed through a feeding tube. By year's end, her medical bills approached $500,000.

In December, the medical staff recommended to Oliver Wanglie, her husband of 53 years, that the ventilator be disconnected. He refused on the grounds that he and his wife believed in the sanctity of life and that, in good conscience, he could never agree to have the ventilator disconnected.

The hospital petitioned the court to have Mr. Wanglie replaced as Helga's legal guardian. The rationale was that Mr. Wanglie had made some statements that indicated that Helga had never voiced an opinion about being sustained in PVS and that he was not legally competent to serve as guardian. The hospital justified its request to disconnect by stating that it did not feel it should be obligated to provide "medically futile" medical treatment.

On July 1, 1991, the judge issued a narrow ruling that the hospital had not demonstrated that Mr. Wanglie was incompetent, and therefore he was the most appropriate legal guardian. Without the tubes being removed, Helga died three days later.

Medical Futility. Although the judge did not address the "medical futility" issue, the concept caught the attention of social scientists, medical ethicists, health care practitioners, and the lay public, and significant attention has now being given to it. It has proven to be an elusive concept to define. Schneiderman, Jecker, and Jonsen have defined **medical futility** as "an expectation of success that is either predictably or empirically so unlikely that its exact probability is often incalculable" (1990:950). In a quantitative sense, "when physicians conclude … that in the last 100 cases, a medical treatment has been useless, they should regard the treatment as futile" (p. 951). In a qualitative sense, "any treatment that merely preserves permanent unconsciousness or that fails to end total dependence on intensive medical care should be regarded as nonbeneficial and, therefore, futile" (p. 952). If a treatment fails to appreciably improve the person as a whole, they argue, physicians are entitled to withhold the treatment without the consent of family or friends.

While courts have supported patients' "negative rights" to refuse medical treatment, the issue of a "positive right" (the request for a particular intervention) is far different. It raises two issues: (1) a resource allocation issue (the more dollars spent on nonbeneficial treatment, the fewer dollars available to spend on those who would benefit), and (2) the reasonableness of requiring health care practitioners to engage in actions they consider to be unwarranted (and possibly harmful). Some believe that unless some benefit is anticipated, specific measures cannot be demanded.

However, successful intervention may be defined differently by different people. Families may be satisfied as long as everything possible is done—even if the patient dies. Their values may rest more on effort than outcome, and they may never define effort as being futile. Some view the issue as showing respect for the autonomy of patients and their families by honoring these value differences but working together to avoid futile treatments (Quill, Arnold, and Back, 2009).

Physician-Assisted Suicide

Physician-assisted suicide occurs when a physician provides a means of death (e.g., a particular drug) and instructions (e.g., how much of the drug would need to be taken for it to be lethal) to a patient but does not actually administer the cause of death. This is different than active euthanasia—a situation in which the physician directly administers the causes of death.

Physician-assisted suicide is legal in some countries in the world (e.g., the Netherlands and Belgium) but not in most. It has become a very controversial issue in the United States where four states—Oregon, Washington, Vermont, and Montana—have legalized the practice. In 1997, the U.S. Supreme Court ruled unanimously that terminally ill persons do not have a constitutional right to physician-assisted suicide, but that states could enact legislation consistent with their own constitution that permits it. Thus far, only the four states have legalized physician-assisted suicide while 39 states have laws prohibiting it. Between 1998 and 2012, 596 individuals in Oregon selected physician-assisted suicide, and 115 did so in Washington in its first three years of legalization. People who engaged in physician-assisted suicide were overwhelmingly white and highly educated, and most had cancer. They mostly feared losing their autonomy, losing the ability to participate in enjoyable activities, and the loss of dignity (O'Reilly, 2010). The box, "Advance Directives and Navajo Culture," describes how this concept is considered in a Native American culture.

Two situations originally drew considerable attention to the issue of physician-assisted suicide in the United States.

Dr. Timothy Quill. In 1991 Dr. Timothy Quill, a 41-year-old general internist in Rochester, New York, published an article in the *New England Journal of Medicine* (1991) in which he described how he assisted "Diane," a 45-year-old woman with leukemia, who had been his patient for eight years, to end her life. Diane had rejected the option of chemotherapy and bone marrow transplantation, which had a one-in-four chance of success, due to the certain negative side effects. After they had a thorough discussion and he was convinced of her full mental competence, Quill prescribed barbiturates and made sure Diane knew how much to take for sleep and how much to take to end her life. They continued to meet regularly, and she promised to meet with him before taking her life—which she ultimately did. While reaction was mixed, Dr. Quill received considerable praise for his action with Diane and his courage in describing the events in print.

Dr. Jack Kevorkian. By far, the key figure in this debate has been **Dr. Jack Kevorkian,** a retired Michigan pathologist who assisted more than 100 people to end their lives by providing a painless means to do so and by being present at the time of death. Each of the persons contacted Kevorkian (none were patients of his) and convinced him that they had

Dr. Jack Kevorkian, a retired Michigan pathologist, repeatedly and successfully challenged prohibitions against physician-assisted suicide. Eventually, he was convicted of second-degree homicide for directly causing the death of an individual who was in the late stages of amyotrophic lateral sclerosis ALS and had requested Kevorkian's help in his death.

made a rational choice to die. The particular means used varied from case to case. Although two juries refused to convict Kevorkian of a crime, ultimately, he engaged in an act of active euthanasia in which he did cause the death to occur. A Michigan jury convicted him in 1999 on this charge and sentenced him to 10 to 25 years in prison. He was released in 2007 after 8 years with a promise that he would no longer assist in suicides, but that he could continue to advocate for physician-assisted suicide. He died in 2011.

Arguments Favoring Physician-Assisted Suicide. Proponents of the legalization of physician-assisted suicide offer the following rationale:

1. It is perfectly appropriate to have physicians and other health care professionals create a comfortable and peaceful environment in which death occurs.
2. People have a right to self-determination; if one has reflected on life circumstances and made a rational, competent decision to die, the assistance of physicians in the act is appropriate. People should not be required to undergo mental and physical decline, endure emotional and physical pain, and incur sizable medical expenses for treatment not desired.
3. In order to prevent abuse, laws could require certain safeguards (e.g., there is intolerable suffering; the patient is mentally competent; a written, witnessed request is provided; the patient consistently and repeatedly over time requests death; and two physicians—one of whom has not participated in the patient's care—agree that death is appropriate).
4. An extremely high rate of suicide already exists. In addition to that which occurs without medical contact, it is clear that many deaths in hospitals occur with some "assistance." An estimated 70 percent of the 1.3 million deaths that occur in American hospitals each year involve some agreement not to take aggressive action to sustain the patient.
5. Public opinion polls show a majority of Americans favor the legalization of physician-assisted suicide and that in certain circumstances as many as half would consider it for themselves. Derek Humphry (1991), president of the Hemlock Society, a national organization in favor of legalized euthanasia, wrote *Final Exit: The Practicalities of Self-Deliverance and Assisted Suicide for the Dying*. It is a how-to-commit suicide guidebook, and it became an overwhelming best seller.

Arguments Opposing Physician-Assisted Suicide. Opponents of physician-assisted suicide offer the following rationale:

1. Traditionally, we have considered the physician's responsibility to be to sustain life and relieve suffering. While actions taken by physicians may sometimes have the opposite effects, their intention is to do good. The primary purpose of physician-assisted suicide, however, is to cause death. The American Medical Association, the American Bar Association, and some medical ethicists believe this to be inconsistent with the physician's professional obligations.
2. Patients considering physician-assisted suicide may be sufficiently ill or so worried that they are not capable of genuine contemplative thought or the exercise of a true informed consent. The depression that might lead to consideration of this act might itself be treatable.
3. The legal possibility of physician-assisted suicide may interfere with a good physician–patient relationship. A physician's willingness to participate may be interpreted by the patient that society (and the physician) would prefer that suicide occur. This interpretation may place implicit pressure on the patient to request the act. Elderly and dying patients may be especially vulnerable psychologically. On a broader scale, a climate may be created in the country in which terminally ill people are expected to end their lives.
4. There could be a slippery-slope argument—by legalizing physician-assisted suicide for patients with terminal illnesses, we increase the likelihood that suicide will become acceptable for other people—the mentally retarded, those with physical disabilities, and the very old, for example.
5. Significant progress has been made in dealing with the pain that often accompanies late-stage diseases. An increasing number of hospitals have developed palliative care programs to manage and control patients' pain, and an increasing number of health care providers are receiving training in palliative care (although there are still far too few programs and trained providers). Hospices are increasingly successful in reducing the amount of pain their clients experience.

Reaction to the two cases mentioned earlier in the general population and among clinicians was divided. The primary point of distinction in the cases is that Dr. Quill assisted in the suicide of a patient that he had known for many years and with whom he had long and engaging discussions about life and death. In that context, he agreed to participate. On the other hand, while Dr. Kevorkian spoke at some length with all his suicides, he did not have long-standing physician–patient interaction with any of them. Many physicians, medical ethicists, and laypersons believe the lack of a personal relationship with the patients is the most troubling aspect of Kevorkian's behavior, although many others believe that he had adequate assurance of the competent desires of each person. He has been both widely praised and widely condemned.

ORGAN DONATION AND TRANSPLANTATION

The ability to successfully transplant organs from a cadaver or living, related donor began with a successful transplant in Boston in 1954 when a 23-year-old man received a kidney from his genetically identical twin brother. The recipient recovered completely and lived another eight years before dying of an unrelated cause. Bone marrow was first successfully transplanted in 1963 (Paris), the same year as the first liver transplant (Denver), the first pancreas was transplanted in 1966 (Minneapolis), the first heart in 1967 (Capetown, South Africa), the first heart–lung in 1981 (Palo Alto, California), the first partial pancreas in 1998 (Minneapolis), the first hand in 1998 (Paris), the first partial face in 2005 (Paris), and the first windpipe in 2008 (Barcelona).

IN COMPARATIVE FOCUS

ADVANCE DIRECTIVES AND NAVAJO CULTURE

Discussion of ethical issues outside the context of particular cultures means that vital information is not considered. On a broad level in the medical context, it increases the likelihood of social policies that conflict with important values of particular groups. On an individual level, it creates potential for misunderstandings and frustration between health care providers and their patients and may lead to suboptimal care.

This is the case with regard to discussions about advance directives with Navajo patients. An important Navajo cultural norm is avoidance of discussion of negative information. In the Navajo belief system, talking about an event increases the likelihood of the event occurring. This belief obviously comes into conflict with several norms in medical settings: discussing risks as well as benefits of intended procedures, telling patients the truth about a negative diagnosis or prognosis, and preparing an advance directive. Ignoring the cultural values of a Navajo patient almost inevitably means that care and treatment will be interrupted and perhaps discontinued.

On the other hand, acting with cultural sensitivity can enable the values of the patient to be respected while medical responsibilities are discharged. Health care providers who work with the Navajo have been encouraged to pursue situations of this type in four ways: (1) determining if the patient is or is not willing to discuss negative information; (2) preparing the patient by building rapport and trust, involving the family, giving an advance warning that bad news is coming, and involving traditional Navajo healers in the encounter; (3) communicating in a kind, caring manner that is respectful of traditional beliefs, for example, referring to the patient in third-party language rather than directly; and (4) following through in a manner that fosters reasonable hope (Carrese and Rhodes, 2000).

Social Policy Issues Related to Organ Transplantation

The success of these sophisticated medical technologies has prolonged the life of many recipients but has also created several complex ethical and social policy issues. Nancy Kutner (1987) and Renee Fox and Judith Swazey (1992) have identified key issues as follows:

1. Do the medical and quality-of-life outcomes for organ recipients (and donors) justify organ transplantation procedures? The question, to be asked before all questions, is whether organ transplants have been shown to have sufficient therapeutic value that they should be continued. If the answer to the first question is affirmative, then at least the four following additional questions need to be addressed.
2. Given that the demand for transplant organs exceeds the supply, how should recipients be selected? Should a complex formula be devised to prioritize potential recipients, or would a first-come–first-served (or random) policy be more consistent with democratic principles?
3. How can the supply of organs be increased? Organ donation policy has evolved during the last four decades, but demand continues to exceed supply. What policy might increase motivation to donate while maintaining a voluntary nature?
4. How much money should be allocated to organ transplant procedures? Given the finite amount of money available to be spent on health care, how much should be directed to these very expensive procedures (most cost between $100,000 and $200,000) that benefit a relatively small number of people versus less dramatic, less costly procedures that might benefit a much larger number?
5. Who should pay for organ transplant procedures? Should (can?) the government be willing to pick up the tab because these

procedures are so expensive? Should health insurance companies guarantee coverage? Today, the federal government's Medicare program pays for kidney, heart, and some liver transplants; some health insurance policies cover transplants; people not covered in these ways are on their own, and those unable to pay are turned away.

This section focuses on the evolution of organ donation policy in the United States, assesses the current and alternative donation policies, and discusses the psychosocial dimension of organ donation.

Organ Donation Policy in the United States

The success of the early transplant efforts in the 1950s and 1960s forced the United States to establish a formal organ donation policy. The initial policy was one of **pure voluntarism**—donation was made legal, and it was hoped that volunteers would come forward. Courts ruled that competent adults could voluntarily donate organs to relatives, which at first seemed like the only possibility. Organs could be donated by minors only with parental and judicial consent.

Donations did increase, but as the supply was increasing, so was the demand. The success rate for transplants improved with the development of the artificial respirator and the heart–lung machine and with the discovery of effective immunosuppressive drugs to suppress the body's immune system, thereby making the recipient's body less likely to reject a transplanted organ.

In 1968, the United States adopted "brain death" as the legal standard for death determination; this definition enabled organs to be taken from those who had suffered irreversible loss of brain function (and therefore were legally dead) but were being sustained on artificial respirators. Death could be pronounced, the organs taken, and then the respirator disconnected.

These changes forced the United States to develop a more assertive organ donation policy: **encouraged voluntarism**. This occurred in 1968 with the passage of the Uniform Anatomical Gift Act (UAGA), which was adopted in every state and Washington, DC by 1971. The UAGA permitted adults to donate all or part of their body after death through donor cards and living wills and gave next of kin authority to donate after an individual's death, as long as no contrary instructions had been given. The UAGA was praised for maintaining the country's allegiance to a voluntary approach and was successful in increasing the number of donors. However, about 90 percent of the 20,000 to 25,000 people who were potential donors each year in the United States still failed to donate; there was no centralized system for identifying those needing and those willing to make a donation; and hospitals had little involvement in the program.

A dramatic change in organ donation policy occurred in 1987 with the establishment of a **weak required request policy**. In order to receive essential Medicare and Medicaid reimbursements, each of the nation's more than 5,000 hospitals must make patients and families aware of the organ donation option and must notify a federally certified organ procurement organization (OPO) when there is consent to donation.

Also, a national central registry (now called the *United Network for Organ Sharing—UNOS*) was created in Richmond, Virginia, to maintain a national list of potential donors and recipients. In order to receive Medicare and Medicaid funding, all the country's more than 260 transplant centers and 58 procurement organizations are required to affiliate with UNOS. When an organ for transplant becomes available, UNOS engages in a matching process considering medical need, medical compatibility (size and blood type), and geographic proximity. Much of the physical work is performed by the OPOs: They encourage donation, contact UNOS when an organ becomes available, send teams to collect and process the organ, and deliver it to its designee.

The weak required request policy maintains the voluntary and altruistic nature of organ donation but seeks to ensure that potential donors are aware of the donation option. The request is sometimes handled very sensitively but sometimes with little enthusiasm and little tactfulness.

Studies indicate that few physicians and nurses have received any education or training in how to make a request for organ donation, and many feel uncomfortable doing so. Many physicians have been unwilling to be the one who asks, citing such reasons as uncertainty about their own attitudes regarding organ donation, lack of knowledge about organ donation criteria and processes, reluctance to bother a family during a time of grief, not knowing how to make the request, and lack of time and reimbursement for the donation request (May, Aulisio, and DeVita, 2000).

Yet studies indicate that more than 90 percent of health care professionals approve of organ donation, and many are willing to donate their own organs. A study of more than 700 physicians in a variety of specialties suggested that organ donation information programs for physicians would alleviate some of their concerns and make them more willing to make the request (McGough and Chopek, 1990).

These attitudes are extremely important because the comfort level of the requestor has been shown to have a significant impact on the likelihood of donation. When families are approached with lines such as, "I don't suppose you want to donate, do you?" or "The law says I have to ask if you want to donate," a refusal is very likely.

Has the 1987 legislation succeeded? The level of criticism directed at UNOS has been minimal; it seems to be well run. (See the accompanying box, "Challenging UNOS.") The number of organ donations did increase after the policy was implemented—but only to a 25 to 30 percent consent level. (It has now increased to 41 percent.) In 2012, more than 28,000 organ transplants (including the heart, liver, kidney, heart–lung, lung, and partial pancreas) occurred—the most ever. Moreover, organ transplants are more likely to be successful than ever before, and recipients are living longer. Yet, more than half of the families that are asked to consider donation after a relative has died do not give their consent.

There are now more than 120,000 people on the waiting lists in Richmond who have been medically and financially approved for transplants. It is likely that two-thirds of these individuals will eventually receive a transplant, but one-third will die before receiving a transplant. The American Council on Transplantation estimates that an additional 100,000 people would benefit from some type of organ transplant but are not on the list because they are unable to demonstrate the means to pay for a transplant. Ironically, research shows that people without health insurance often donate organs but rarely receive them (Herring, Woolhandler, and Himmelstein, 2008).

Alternative Directions for Organ Donation Policy

There is some support in the United States for adopting a different or enhanced policy regarding organ donation. A recent study of transplant surgeons, coordinators, and nurses found that the current policy of altruistic donation was rated as being the most morally appropriate policy, but that several alternative, more aggressive policies were also considered to be morally appropriate (Jasper et al., 2004). Should the United States again choose to revise its organ donation policy, five main alternatives exist:

1. **Strong required request.** Every citizen would be asked to indicate her or his willingness to participate in organ donation—either on income tax returns or through a mandatory check-off on the driver's license. This policy retains the voluntary/altruistic nature of the system but is more aggressive in forcing people to consider organ donation and to take a formal position. Some states already do this, and many have more than a 50 percent donor rate (Alaska and Montana are highest at 76 percent followed by Utah and Washington at 72 percent). Most states do not, and several have less than a 30 percent donor rate (Texas is the lowest at 7 percent followed by New York at 15 percent and South Carolina at 16 percent).

IN THE FIELD

CHALLENGING UNOS

While UNOS has generally been praised for its commitment to a fair allocation system, there are occasions when the anguish of a particular individual on the waiting list creates a challenge for the organization's protocols. Such was the situation in 2013 with the simultaneous cases of 10-year-old Sarah Murnaghan of Newtown Square, Pennsylvania and 11-year-old Javier Acosta of New York City. Both children had end-stage cystic fibrosis, were terminally ill, and required a lung transplant to extend their lives. More than 80 percent of cystic fibrosis patients who receive a lung transplant survive at least a year, and 50 percent survive at least five years.

However, UNOS policy requires that patients under age 12 receive a lung only from another adolescent. The policy was developed because lungs from an adult, which become available much more often, would be too large for a child-size body. As Sarah's prognosis grew worse, her parents made public pleas for reconsideration of the policy. Politicians got involved and requested that the Secretary of Health and Human Services, Kathleen Sebelius (whose cabinet department oversees UNOS) mandate a lung transplant for Sarah. She declined with the reasoning that the protocols were set up in the fairest way possible and were time-tested. A lawsuit was filed, and the judge ruled the next day that Sarah should be moved to the front of the queue until a formal court hearing could occur a few days later.

This prompted a special review board at UNOS to consider the case. They recommended no change in the policy saying it would be wrong to hastily make a change based on one or two cases and that any policy that grants an organ to one person necessarily deprives another from receiving the organ. However, they did advise that a special appeals process be created for situations like Sarah's. Before that hearing could occur, an adult lung became available, Sarah was first on the list, and she received a double-lung transplant. Almost immediately, Sarah had difficulties, and doctors had a difficult time weaning her from the respirator. Several days later, the double-lungs were removed, but a new double-lung set was transplanted. Javier Acosta had also been added to the adult list and was waiting to be selected for a transplant.

2. **Weak presumed consent.** Hospitals would be required by law to remove and use all suitable cadaver organs for donation unless the deceased had expressly objected, through a central registry, a nondonor card, or if family members objected. Sometimes called "routine salvaging of organs," this remains a voluntary/altruistic system but with a very significant change—the presumption or default position is that donation will occur. In presumed consent, the person must take an action to prevent donation. Several European countries have this policy.
3. **Strong presumed consent.** Physicians would be given complete authorization to remove usable organs regardless of the wishes of the deceased or family members. Also referred to as "expropriation," it is the only alternative that eliminates the voluntary dimension of organ donation policy, but it is also the policy that retrieves the largest number of transplantable organs. This policy also is in effect in several European countries.
4. **Weak market approach.** Individuals or next of kin for deceased donors would receive a tax benefit for the donation of organs (perhaps a one-year deduction from federal and state taxes) or a cash payment of sufficient size to offset some funeral expenses. This approach adds financial incentive to a volunteer-based system, thus reducing the role of altruism. Proponents argue that offering financial benefit is fair and sensitive, but opponents prefer altruistic motivation. They worry that

some likely donors would be so offended by the suggestion of payment that they would choose not to donate. Pennsylvania became the first state to incorporate this technique in 2000, when it began offering $300 to help families of organ donors cover their funeral expenses.

5. **Strong market approach.** Individuals or next of kin would be able to auction organs to the highest bidder. Like the weak market approach, this system places greater emphasis on increasing the number of organs donated than on retaining altruistic motivation for donation. Critics of the strong market approach worry that the financial incentive may place undue pressure on low-income persons and people in the Third World to donate. Once organs are on the market, the wealthy clearly would have easier access. The buying and selling of organs is unlawful in the United States at this time but is an option in some European countries. A former company in Germany routinely sent a form letter to all persons listed in the newspaper as having declared bankruptcy. The individual was offered $45,000 (plus expenses) for a kidney (which was then sold for $85,000). In 1995, India, the country in which the most people have sold organs to strangers, banned the practice and now restricts living donor-related transplants to relatives.

The Psychosocial Dimension of Organ Transplantation and Donation

Some of the most insightful work done in medical sociology and medical anthropology has pertained to attitudes, motivations, and consequent feelings related to organ donation. Anthropologist Lesley Sharp (2006) has recently written about not only the illuminated side of organ transplantation—the medical successes and altruistic motivations—but also the less studied complicated relationships and social injustices related to the gap between supply and demand. Renee Fox and Judith Swazey (1978, 1992), leaders for many years in this field of study in medical sociology, have written forcefully that the organ donation decision needs to be placed within a social-structural context. Based on years of systematic observation in transplantation settings and countless interviews with physicians, patients, donors, and families, they have raised three important concerns with donation and transplantation:

1. There is still too much "uncertainty" about the therapeutic value of organ transplantation. While significant progress has been made, there are still concerns. For example, there is still a high incidence of long-term (five or ten years posttransplant) or chronic rejection of transplanted organs; many organ recipients are prone to redeveloping the same life-threatening medical conditions that led to the transplant; and there have been repeated failures with certain transplant modalities (e.g., animal to human transplants and the totally implantable artificial heart). Fox and Swazey argue that these types of questions have not been given adequate and genuine reflection.
2. There has been too little consideration of the psychosocial dimensions of organ donation and receipt. The focus of attention has become so fixed on the "organ shortage problem" and the "allocation of scarce resources" that the human dimensions of the process have seemed inconsequential. The concept of the "gift-exchange" relationship between donor and recipient has given way to discussions of "supply and demand" and compensation for donors. Fox and Swazey contend that this way of thinking commodifies body parts and reconceptualizes the "gift of life" idea.
3. The high price of transplant procedures and the fact that more people seek an organ than have been willing to donate one have created real, and not yet fully answered, questions of distributive justice and public good.

The Donor–Recipient Relationship

Although research has shown that the decision to donate a kidney to a relative is typically made very quickly and with little regret, the decision to donate and the decision to receive an organ is governed by often unspoken but powerful social norms. Family members may express an "intense desire" to make a potentially lifesaving gift to someone close, and they sometimes feel that donating is part of a family obligation (even sensing some family pressures). With this "gift of life," an incredible bond can be established between the giver and the receiver. In addition, some cases of "black sheep donors" have surfaced, in which individuals who felt remorse for previous wrongs against the family wished to make amends through donation of an organ (Fox, 1989). For those who donate to strangers, there is often enhancement of self-image—a feeling of having actualized traits of helpfulness and generosity to others and having demonstrated their own altruistic nature.

Just as prevailing norms may motivate organ donation, they also motivate accepting it. Rejection of an offer to donate constitutes a form of rejection of the donor. When the donation and transplant occur, a type of "obligation to repay" is incurred by the recipient. Having received something so profoundly important, the recipient becomes, in a sense, a debtor—owing something back to the donor (Shaw, 2010). Fox and Swazey (1978) referred to this as the **tyranny of the gift**.

Similar considerations occur in family members' decisions to offer cadaver organs. These decisions are almost always made in traumatic situations, such as automobile accidents, in which the death is sudden and there is no preparation for it. Family members often consent to donation as a way of bringing meaning or some sense of value into a senseless tragedy. The altruistic and humanitarian aspects of organ donation become powerful motivating forces (Fox and Swazey, 1978).

Fox and Swazey's *Spare Parts: Organ Replacement in American Society* (1992) is a powerful critique of organ donation and organ transplantation in American society. They indict society for the extent to which it has become obsessed with rebuilding people and sustaining life at all costs and for doing so while failing to fully consider quality of life considerations and the psychosocial aspects of donation and transplantation. The final chapter of their book describes their decision to leave this entire field of inquiry.

> By our leave-taking we are intentionally separating ourselves from what we believe has become an overly zealous medical and societal commitment to the endless perpetuation of life and to repairing and rebuilding people through organ replacement—and from the human suffering and the social, cultural, and spiritual harm we believe such unexamined excess can, and already has, brought in its wake. (1992:210)

ASSISTED PROCREATION

Infertility

The reported incidence of **infertility**, defined as the absence of pregnancy after one year of regular sexual intercourse without contraception, is increasing in the United States. An estimated 10 to 15 percent of American couples of childbearing age are defined as being infertile. However, this percentage may give an exaggerated picture; up to half of the couples not pregnant after one year get pregnant on their own in the second year. The increased incidence of infertility is due to several factors: an actual increase in infertility (likely due to increased exposure to radiation and pollution, ingesting of certain drugs, and higher levels of sexually transmitted infections); the number of older women now trying to get pregnant (peak fertility occurs between 20 and 29); and more couples seeking assistance with lack of fertility.

The specific reason for a couple's infertility is traceable to the female partner in approximately 40 percent of the cases. The most common specific causes are inability to produce eggs for fertilization, blocked fallopian tubes (so that the eggs cannot travel to meet the sperm), and a sufficiently high level of acidity in the vagina that kills deposited sperm. In another 40 percent of infertility cases, the specific cause is traceable to the male partner—typically, low sperm count and/or low sperm motility. The specific